COURSE

Cambridge IGCSE Business (0450 / 0264)

Production and capacity

4 Operations management · 4.1 Production of goods and services: productivity

Revision for 4 Operations management: each firm works it out on its own deals.

Class	Topic
Cambridge IGCSE Business (0450 / 0264)	**Production and capacity**

THE LESSON

Setup					2 min
Round 1					Decisions 3 min · Deals 5 min · Topic tasks 4 min
Firm decisions	**Deals**	TASK 1 **Cars per staff**	TASK 2 **Raise output**		
Round 2					Decisions 3 min · Deals 5 min · Topic tasks 4 min · B2B deal 2 min
Firm decisions	**Deals**	TASK 3 **Rank productivity**	TASK 4 **New productivity**	**B2B deal**	
Round 3					Decisions 3 min · Deals 5 min · Topic tasks 4 min · B2B deal 2 min
Firm decisions	**Deals**	TASK 5 **One-off big order**	TASK 6 **Productivity rise %**	**B2B deal**	
Final questions					3 min

45-minute setting shown. The 30-minute setting has one topic task per round (Tasks 3, 5 and 6) and one B2B deal. 60 and 75 add more deals.

Each step

Firm decisions	Each role makes its decision on its own page.
Deals	Each pupil trades with a classmate in another firm.
Topic task	The firm answers together. Everyone presses Agree.
B2B deal	One firm sells a lot of cars to another firm.
Final questions	Each pupil answers six questions.

Firm decisions

CEO	**Staff**	Hire staff. Each costs a salary and gives one benefit.
	Sabotage FROM ROUND 2	Hit a rival firm's cash, at a cost. A Legal Counsel blocks it.
Sales & Marketing	**Marketing**	Book one advert campaign to lift the next sales.
Finance	**Capital & Loans**	Borrow cash. The firm pays it back with interest.
	Investments	Buy property or funds. The value can rise or fall.
Operations	**Buying**	Hire the Procurement Specialist or the Stock Controller for the round.

B2B deal · Rounds 2 and 3

One firm sells a lot of 3 cars to another firm (4 in Round 3). With an odd number of firms, two firms offer to sell and the lower price wins.

Topic tasks

Each firm gets its own figures. The method is the same for every firm.

Task 1

ROUND 1

The prep workshop.

Cars prepared per member of staff a day?

WORK OUT THE NUMBER · EXAMPLE FIGURES

Cars prepared a week	**60 cars**
Staff	**6 staff**
Days worked a week	**5 days**

METHOD

Per staff a day = cars a week ÷ staff ÷ days

60 ÷ 6 = 10 a week; ÷ 5 = 2 cars

Why: Productivity is output for each worker in a set time.

Task 2

ROUND 1

Your prep workshop is slow.

Which two would raise output per worker?

PICK TWO

- ✔ **Buy better equipment**
- ○ Cut the hours staff work
- ○ Hire more staff with the same tools
- ○ Raise your prices
- ✔ **Train staff in faster methods**

Why: Productivity is output per worker; more staff raise output, not productivity.

Task 3

ROUND 2

Three workshops: cars prepared a week, and the staff who prepare them.

Order by productivity, highest first?

PUT THESE IN ORDER, FIRST TO LAST · RIGHT ORDER

1. **24 cars, 4 staff**
2. **40 cars, 8 staff**
3. **45 cars, 15 staff**

Why: Cars per worker: 6, 5 and 3. The busiest workshop is the least productive.

Task 4

ROUND 2

More staff, more cars.

New cars prepared per member of staff?

WORK OUT THE NUMBER, THEN PICK IT · EXAMPLE FIGURES

Cars prepared a week	**28 cars**
Staff	**4 staff**
Extra staff	**3 staff**
Extra cars prepared a week	**28 cars**

METHOD

Per staff = (cars + extra cars) ÷ (staff + extra staff)

(28 + 28) ÷ (4 + 3) = 8 cars

Why: Productivity rose from 7 to 8 cars per worker.

Task 5

ROUND 3

Your workshop is fully booked. A one-off big order arrives.

How should you handle it, and why?

PICK A OR B AND GIVE A REASON

- ○ Build a new prep bay
- ✔ **Pay overtime for this order**

Why: A one-off order does not justify the cost of a new bay.

Task 6

ROUND 3

New tools and training.

Rise in cars per worker, as a %?

WORK OUT THE NUMBER · EXAMPLE FIGURES

Cars a week, before	**16 cars**
Staff, before	**4 staff**
Cars a week, after	**18 cars**
Staff, after	**3 staff**

METHOD

Rise % = (cars per worker after − before) ÷ before × 100

16 ÷ 4 = 4; 18 ÷ 3 = 6; (6 − 4) ÷ 4 × 100 = 50%

Why: Each worker now prepares 6 cars a week, not 4.

Final questions

6
questions for each pupil

159
Level 2 Business questions

6
of them on production and capacity

The game draws 6 questions on this topic.

1 Which situation is best suited to job production?
Restoring one classic car to a customer's wishes.

2 A parts maker produces twenty identical service kits, then resets the equipment to make a different kit. Which production method is this?
Batch production.

3 What is a likely advantage of flow production for standardised components?
High volumes at a low cost per unit.

4 A workshop queues every Saturday but has spare weekday capacity. What should it do before paying for a permanent extra bay?
First test moving work to weekdays.

5 A viral review causes a sudden rise in service bookings. What is the best short-term response?
Prioritise bookings and add trained help.

6 Labour productivity = total output ÷ number of employees. Example: 400 services ÷ 10 technicians = 40 per technician. Now 540 services and 12 technicians: what is labour productivity?
45 services per technician.